Predicting Coronary Heart Disease from Lifestyle Survey Data with Weighted Logistic Regression

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***Abstract*—We predict whether a person has been diagnosed with coronary heart disease (MICHD) from 321 answers to the BRFSS health survey. The data is noisy (missing values, special answer codes) and strongly imbalanced (8.8% of positive cases), so that the six basic methods seen in class reach an F1 score of at most 0.11. We decode the survey answer codes, one-hot encode the categorical answers, add pairwise interactions between the most informative features, and train an L2-regularized, class-weighted logistic regression with Newton's method, whose regularization and decision threshold are tuned by cross-validation to maximize the F1 score. The final model reaches F1 = 0.443 (accuracy 0.882) on the AIcrowd test set. An ablation study shows that adapting the decision rule to the class imbalance is by far the most important ingredient.**

I. Introduction

Cardiovascular diseases are a leading cause of death worldwide. The Behavioral Risk Factor Surveillance System (BRFSS) is a telephone survey of U.S. residents about their health and lifestyle. Given the answers of a person, we predict $y = 1$ if the person has MICHD (heart attack or angina) and $y = -1$ otherwise. The training set contains $N = 328{,}135$ people and $D = 321$ features, the test set 109,379 people. Since only 8.8% of the labels are positive, always predicting -1 gives 91% accuracy but is useless; we therefore optimize and report the F1 score of the positive class, the metric of the competition. Only NumPy is used.

II. Models and Methods

A. Evaluation protocol

The training set is randomly split (seed 42) into a 60% *training part* and a 40% *hold-out part*. The preprocessing and the hyperparameters are learned on the training part only (hyperparameters by 5-fold cross-validation, CV), and Tables I and II are measured on the hold-out part. The final model is retrained on the full training set (`run.py`).

B. Preprocessing

The pipeline (`preprocessing.py`) is fitted on the training data and applied unchanged to the other sets:

1) drop the 16 survey metadata features (interview date, sample ids, sampling weights), which do not describe the person;
2) *decode the BRFSS answer codes*, with a rule deduced from the largest value of each feature: 7/9, 77/99, 777/999 ("don't know", "refused") become missing; 8/88/888 ("none") and 555 ("never") become 0; frequencies coded `Pxx` (e.g. 203 = 3 times a week) are converted to times per day, which matches the variables pre-computed by the BRFSS (correlation 0.9999);
3) drop the features with $\geq 95\%$ missing values and the almost constant ones (one value in $\geq 99\%$ of the non-missing entries);
4) *one-hot encode* the non-binary features with at most 10 values (missing counted as a value), with a reference category;
5) standardize the other features and replace missing values by 0 (the mean); drop features with correlation ≥ 0.99 with a previous one;
6) *interactions*: add the standardized products of all pairs among the 30 features most correlated with the label (age, general health, diabetes, high blood pressure, stroke, ...), removing constant products.

This gives $D' = 866$ features (441 without interactions).

C. Baselines

We train the six methods of `implementations.py` on the cleaned features without interactions (some products are collinear, which makes least squares singular): linear regression by gradient descent (GD, 100 steps, $\gamma = 0.02$), by stochastic gradient descent (SGD, 10^4 steps, $\gamma = 10^{-3}$), by the normal equations (least squares) and with an L2 penalty (ridge, $\lambda = 4.6 \cdot 10^{-4}$ chosen by 5-fold CV on the RMSE), and logistic regression without and with penalty ($\lambda = 10^{-4}$) by gradient descent (100 steps, $\gamma = 0.1$, labels in $\{0, 1\}$). The step sizes are below the stability limits $2/L$ and $8/L$, with $L \approx 49$ the largest eigenvalue of $X^\top X/N$. All predict 1 if $\mathbf{x}^\top \mathbf{w} \geq 0$.

D. Our model

We minimize a class-weighted, L2-regularized logistic loss

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^{N} c_n \log\left(1 + e^{-y_n \mathbf{x}_n^\top \mathbf{w}}\right) + \lambda \|\mathbf{w}\|^2, \qquad (1)$$

with balanced weights $c_n = N/(2N_{y_n})$ (N_{y_n}: size of the class of n), so that both classes have the same total weight. We use *Newton's method*: each step solves $H\,\Delta\mathbf{w} = \nabla\mathcal{L}$ with $H = \frac{1}{N} X^\top S X + 2\lambda I$, $S = \mathrm{diag}(c_n \sigma_n (1 - \sigma_n))$, $\sigma_n =$

Table I
THE SIX BASIC METHODS ON THE HOLD-OUT SET (THRESHOLD 0, CLEANED FEATURES WITHOUT INTERACTIONS).

Method	F1	Accuracy
Least squares GD	0.000	0.911
Least squares SGD	0.006	0.911
Least squares (normal equations)	0.106	0.913
Ridge regression	0.100	0.913
Logistic regression (GD)	0.063	0.911
Reg. logistic regression (GD)	0.062	0.911

Table II
ABLATION STUDY ON THE HOLD-OUT SET. NEWTON LOGISTIC REGRESSION, THRESHOLD t TUNED BY 5-FOLD CV UNLESS STATED OTHERWISE.

Variant	D'	t	F1	Acc.
Basic preprocessing*	417	0.9	0.433	0.866
+ metadata, codes decoded	332	1.0	0.434	0.874
+ one-hot up to 10 values	441	0.9	0.433	0.866
+ interactions (final)	866	1.1	0.432	0.878
Final, no class weights	866	-1.3	0.435	0.876
Final, threshold $t = 0$	866	0	0.378	0.760

* no decoding, one-hot up to 5 values, $\lambda = 2.15 \cdot 10^{-5}$ (our first submission).

$\sigma(\mathbf{x}_n^\top \mathbf{w})$. It converges in about 10 steps and needs no step size. A sample is classified as positive if $\mathbf{x}^\top \mathbf{w} \geq t$.

The regularization $\lambda \in \{10^{-4}, 10^{-3.5}, \ldots, 10^{-2}\}$ and the threshold $t \in \{-2, -1.9, \ldots, 1.9\}$ are chosen *together* by 5-fold CV on the training part, maximizing the mean validation F1 score: $\lambda = 10^{-3}$ and $t = 1.1$ (CV F1 = 0.425, Fig. 1). The final model uses 15 Newton steps.

III. RESULTS

Table I shows that all basic methods reach about 91% accuracy, the accuracy of always predicting -1: because of the imbalance, almost all scores are negative and the F1 score stays below 0.11. GD and SGD are even worse than the closed-form solution, as they are far from convergence with a stable step size.

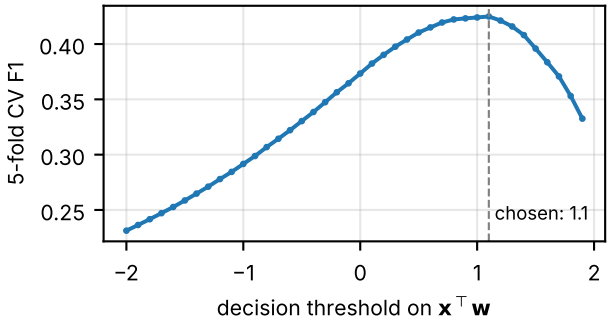


Figure 1. Mean 5-fold CV F1 score of the final model ($\lambda = 10^{-3}$) as a function of the decision threshold t.

Table II shows that the decision rule is what matters: with the default threshold the final model predicts 30% of positives and reaches only 0.378, while the tuned threshold gives 0.432 (+0.055). Class weights and threshold tuning are interchangeable: without weights, the tuned threshold ($t = -1.3$) gives the same F1 score.

The differences between the preprocessing variants on the hold-out set (0.432–0.434) are smaller than its noise ($\approx \pm 0.004$ with 11,500 positive samples). We therefore also compared the first and the final pipeline with a paired 5-fold CV on the full training set (same folds, best threshold of each fold): the final pipeline is better on all 5 folds, by $+0.004$ F1 on average ($+0.002$ without the interactions). The test set confirms it: our first submission obtained F1 = 0.438 (accuracy 0.869) on AIcrowd, the final model **F1 = 0.443** (accuracy 0.882).

IV. DISCUSSION

The main difficulty of this dataset is the class imbalance, not the model complexity: a linear model reaches an F1 score of 0.43 once its decision rule is adapted, and better features only add a few thousandths. We also tried quantile binning of the continuous features (any non-linear shape per feature), clipping of outliers, missing-value indicators and one-hot encoding up to 15 values, without improvement; only the interactions, which model joint effects (e.g. age with diabetes), helped slightly. Decoding the answer codes mainly makes the features meaningful and the model smaller: most special codes were already captured by one-hot encoding.

The model has limits. The threshold optimizes F1, which favors recall over accuracy; a medical use could prefer another trade-off. The preprocessing, including the choice of the interactions, is fitted once on the training part before cross-validation, which slightly leaks information into the validation folds; the agreement between CV, hold-out and test scores suggests that this effect is small.

V. SUMMARY

Decoding the survey answers, one-hot encoding, pairwise interactions, and a weighted L2-regularized logistic regression trained with Newton's method, with its regularization and threshold tuned by cross-validation for the F1 score, raise the F1 score from at most 0.11 for the basic methods to 0.443 on the test set. Handling the class imbalance accounts for most of the gain. All results are reproduced by `run.py` and `exploration.ipynb`.